

Brutal Practice Paper B53 - Transportation, Large Numbers, Winds & Storms, Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. An athlete's heart beats 72 times each minute at rest. During a race the rate increases by 25 percent. How many times does her heart beat during a 4-minute race? [\[Transportation in Animals & Plants\]](#)
 - (A) 450 beats
 - (B) 90 beats
 - (C) 360 beats
 - (D) 306 beats
 - (E) 270 beats
 - (F) 288 beats
 - (G) 348 beats

2. Write the number 'three crore five lakh forty thousand two hundred' in figures. [\[Large Numbers\]](#)
 - (A) 3,50,40,200
 - (B) 30,54,200
 - (C) 30,50,40,200
 - (D) 3,00,54,200
 - (E) 3,05,40,200
 - (F) 3,05,40,020

3. A cyclone's winds blow at 180 km per hour. How many metres does the wind travel in just 1 second? [\[Winds, Storms & Cyclones\]](#)
 - (A) 50 metres
 - (B) 3 metres
 - (C) 500 metres
 - (D) 60 metres
 - (E) 30 metres
 - (F) 5 metres
 - (G) 18 metres

4. A rectangular garden measures 40 m by 25 m. A path 2 m wide is laid all around the outside of it. What is the area of the path alone? [\[Perimeter & Area\]](#)
 - (A) 276 square m
 - (B) 252 square m
 - (C) 300 square m
 - (D) 1,276 square m
 - (E) 130 square m
 - (F) 268 square m

5. Human blood is about 55 percent plasma by volume. A man has 5 litres (5000 mL) of blood. How many millilitres of his blood are NOT plasma? [\[Transportation in Animals & Plants\]](#)
- (A) 1250 mL
 - (B) 2500 mL
 - (C) 225 mL
 - (D) 550 mL
 - (E) 4500 mL
 - (F) 2250 mL
6. In the number 7,80,46,295, what is the place value of the digit 8? [\[Large Numbers\]](#)
- (A) 8,00,000
 - (B) 800
 - (C) 8
 - (D) 8,00,00,000
 - (E) 80,000
 - (F) 80,00,000
7. The spinning cups of an anemometer turn 1,500 times in 5 minutes. How many turns is that per second? [\[Winds, Storms & Cyclones\]](#)
- (A) 25 turns per second
 - (B) 30 turns per second
 - (C) 300 turns per second
 - (D) 75 turns per second
 - (E) 5 turns per second
 - (F) 0.2 turns per second
 - (G) 50 turns per second
8. A triangle has a base of 18 cm and a height of 12 cm. What is its area? [\[Perimeter & Area\]](#)
- (A) 120 square cm
 - (B) 30 square cm
 - (C) 54 square cm
 - (D) 216 square cm
 - (E) 100 square cm
 - (F) 96 square cm
 - (G) 108 square cm
9. A plant loses 250 mL of water by transpiration on a normal day. On a hot, windy day it loses 40 percent more. How much water does it lose on the hot day? [\[Transportation in Animals & Plants\]](#)
- (A) 290 mL
 - (B) 100 mL
 - (C) 150 mL
 - (D) 450 mL
 - (E) 350 mL
 - (F) 210 mL
 - (G) 360 mL

10. Round the number 4,86,729 to the nearest ten thousand. [\[Large Numbers\]](#)

- (A) 4,80,000
- (B) 4,87,000
- (C) 49,00,000
- (D) 5,00,000
- (E) 4,90,000
- (F) 4,90,729
- (G) 4,86,000

11. When air over a hot city expands, becomes lighter and rises, it leaves low pressure below. From where does fresh air then rush in to take its place? [\[Winds, Storms & Cyclones\]](#)

- (A) From high in the sky straight downward only
- (B) No air moves because the pressure is equal everywhere
- (C) From the low-pressure city toward an even lower area
- (D) From warm regions toward warmer regions
- (E) From the city outward to the surroundings
- (F) From the cooler, higher-pressure surroundings toward the city

12. A circular running track has a radius of 35 m. Using pi as $\frac{22}{7}$, find the distance once around the track.

[\[Perimeter & Area\]](#)

- (A) 70 m
- (B) 110 m
- (C) 220 m
- (D) 3,850 m
- (E) 176 m
- (F) 440 m

13. A nurse counts 18 pulse beats in 15 seconds. Since the pulse matches the heartbeat, what is this person's heart rate per minute? [\[Transportation in Animals & Plants\]](#)

- (A) 18 beats per minute
- (B) 36 beats per minute
- (C) 60 beats per minute
- (D) 45 beats per minute
- (E) 270 beats per minute
- (F) 72 beats per minute

14. The Earth is about 14,96,00,000 km from the Sun and 3,84,000 km from the Moon. How much farther from Earth is the Sun than the Moon? [\[Large Numbers\]](#)

- (A) 1,49,21,600 km
- (B) 15,00,00,000 km
- (C) 14,92,160 km
- (D) 14,92,16,000 km
- (E) 14,57,60,000 km
- (F) 14,99,84,000 km
- (G) 14,96,00,000 km

- 15.** During a thunderstorm you see a lightning flash and hear the thunder 6 seconds later. If sound travels 330 metres per second, about how far away is the storm? [\[Winds, Storms & Cyclones\]](#)
- (A) 55 metres
 - (B) 198 metres
 - (C) 1,800 metres
 - (D) 1,980 metres
 - (E) 660 metres
 - (F) 19,800 metres
- 16.** Find the area of a circle whose radius is 14 cm. Use pi as $\frac{22}{7}$. [\[Perimeter & Area\]](#)
- (A) 154 square cm
 - (B) 616 square cm
 - (C) 308 square cm
 - (D) 440 square cm
 - (E) 196 square cm
 - (F) 1,232 square cm
- 17.** If a heart beats 72 times every minute without stopping, about how many times does it beat in one full day (1 day = 1440 minutes)? [\[Transportation in Animals & Plants\]](#)
- (A) 1036800 beats
 - (B) 4320 beats
 - (C) 17280 beats
 - (D) 86400 beats
 - (E) 103680 beats
 - (F) 10368 beats
- 18.** A factory makes 2,450 toys every day. How many toys does it make in a non-leap year of 365 days? [\[Large Numbers\]](#)
- (A) 89,425 toys
 - (B) 8,94,000 toys
 - (C) 7,35,000 toys
 - (D) 8,94,250 toys
 - (E) 89,42,500 toys
 - (F) 9,04,250 toys
 - (G) 8,82,000 toys
- 19.** A cyclone is roughly a circle of diameter 400 km. Using pi as about 3.14, estimate the distance once around its outer edge. [\[Winds, Storms & Cyclones\]](#)
- (A) 1,256 km
 - (B) 1,000 km
 - (C) 2,512 km
 - (D) 1,200 km
 - (E) 400 km
 - (F) 5,02,400 km
 - (G) 628 km

20. A square has a perimeter of 56 cm. What is the area of the square? [\[Perimeter & Area\]](#)

- (A) 784 square cm
- (B) 112 square cm
- (C) 224 square cm
- (D) 49 square cm
- (E) 56 square cm
- (F) 144 square cm
- (G) 196 square cm

21. The xylem tubes of a tall tree carry 18 litres of water up the trunk each day. How many litres rise through the tree during the whole month of June (30 days)? [\[Transportation in Animals & Plants\]](#)

- (A) 540 litres
- (B) 480 litres
- (C) 18 litres
- (D) 510 litres
- (E) 5400 litres
- (F) 48 litres
- (G) 600 litres

22. Write the number 9,80,75,310 using the International (comma) system of grouping. [\[Large Numbers\]](#)

- (A) 9,80,75,310
- (B) 9,807,5310
- (C) 98,750,310
- (D) 98,075,310
- (E) 980,753,10
- (F) 980,75,310

23. A cyclone dumps 5 cm of rain evenly over a flat field measuring 200 m by 150 m. What volume of water, in cubic metres, falls on the field? [\[Winds, Storms & Cyclones\]](#)

- (A) 3,000 cubic metres
- (B) 1,500 cubic metres
- (C) 300 cubic metres
- (D) 30,000 cubic metres
- (E) 15,00,000 cubic metres
- (F) 15,000 cubic metres
- (G) 150 cubic metres

24. A rectangular field measures 120 m by 90 m. Fencing costs Rs 25 for each metre. What is the total cost to fence the whole field? [\[Perimeter & Area\]](#)

- (A) Rs 1,05,000
- (B) Rs 5,250
- (C) Rs 2,70,000
- (D) Rs 21,000
- (E) Rs 9,500
- (F) Rs 10,500

- 25.** A monitor records that a heart took exactly 200 beats. If the heart was beating at 80 beats per minute, how many seconds did those 200 beats take? [\[Transportation in Animals & Plants\]](#)
- (A) 2.5 seconds
 - (B) 16000 seconds
 - (C) 75 seconds
 - (D) 160 seconds
 - (E) 150 seconds
 - (F) 120 seconds
- 26.** What is the predecessor of the smallest 8-digit number? [\[Large Numbers\]](#)
- (A) 1,00,00,001
 - (B) 99,99,999
 - (C) 99,99,990
 - (D) 9,99,99,999
 - (E) 99,99,998
 - (F) 10,00,000
 - (G) 1,00,00,000
- 27.** A storm's wind blows at 25 metres per second. What is this wind speed in kilometres per hour? [\[Winds, Storms & Cyclones\]](#)
- (A) 25 km per hour
 - (B) 9 km per hour
 - (C) 150 km per hour
 - (D) 45 km per hour
 - (E) 90 km per hour
 - (F) 900 km per hour
 - (G) 6.94 km per hour
- 28.** A parallelogram has a base of 25 cm and a height of 8 cm. What is its area? [\[Perimeter & Area\]](#)
- (A) 33 square cm
 - (B) 50 square cm
 - (C) 100 square cm
 - (D) 400 square cm
 - (E) 160 square cm
 - (F) 200 square cm
 - (G) 66 square cm
- 29.** Healthy blood contains about 5,000,000 red blood cells in each cubic millimetre. Which is the correct way to write 5,000,000 using a power of ten? [\[Transportation in Animals & Plants\]](#)
- (A) 5×10 to the power 6
 - (B) 5×10 to the power 5
 - (C) 5×10 to the power 3
 - (D) 50×10 to the power 4
 - (E) 6×10 to the power 5
 - (F) 5×10 to the power 7
 - (G) 0.5×10 to the power 6

30. Find the sum of the greatest 5-digit number and the smallest 6-digit number. [\[Large Numbers\]](#)

- (A) 2,00,001
- (B) 10,99,999
- (C) 2,00,000
- (D) 1,98,999
- (E) 1,99,999
- (F) 1,99,998
- (G) 1,99,990

31. Air pressure falls by about 12 hectopascals for every 100 metres you climb. By how much does the pressure fall when you climb 850 metres? [\[Winds, Storms & Cyclones\]](#)

- (A) 12 hectopascals
- (B) 102 hectopascals
- (C) 120 hectopascals
- (D) 100 hectopascals
- (E) 84 hectopascals
- (F) 1,020 hectopascals
- (G) 96 hectopascals

32. A floor measures 6 m by 4 m and is to be covered with square tiles of side 20 cm. How many tiles are needed? [\[Perimeter & Area\]](#)

- (A) 6,000 tiles
- (B) 600 tiles
- (C) 60 tiles
- (D) 2,40,000 tiles
- (E) 150 tiles
- (F) 1,200 tiles
- (G) 300 tiles

33. Each heartbeat pumps about 70 mL of blood out of the heart. At 72 beats per minute, how many litres of blood does the heart pump in one minute? [\[Transportation in Animals & Plants\]](#)

- (A) 0.504 litres
- (B) 5.4 litres
- (C) 50.4 litres
- (D) 504 litres
- (E) 5.04 litres
- (F) 5040 litres
- (G) 142 litres

34. Which number is equal to 5×10^7 plus 3×10^5 plus 4×10^2 plus 9? [\[Large Numbers\]](#)

- (A) 53,00,409
- (B) 5,03,40,009
- (C) 5,03,00,409
- (D) 5,30,00,409
- (E) 5,03,00,049
- (F) 5,00,34,009
- (G) 50,30,00,409

35. A cyclone is moving straight toward a coast that is 270 km away, travelling at 15 km per hour. How many hours until it reaches the coast? [\[Winds, Storms & Cyclones\]](#)

- (A) 12 hours
- (B) 27 hours
- (C) 15 hours
- (D) 36 hours
- (E) 18 hours
- (F) 4,050 hours
- (G) 255 hours

36. An L-shaped plot is made from a 10 m by 6 m rectangle with a 4 m by 3 m corner piece removed. What is the area of the L-shaped plot? [\[Perimeter & Area\]](#)

- (A) 66 square m
- (B) 60 square m
- (C) 72 square m
- (D) 42 square m
- (E) 54 square m
- (F) 48 square m
- (G) 12 square m

37. A leaf has 500 stomata, and each stoma lets out about 0.02 mL of water vapour in an hour. How much water does the whole leaf lose by transpiration in 5 hours? [\[Transportation in Animals & Plants\]](#)

- (A) 5 mL
- (B) 100 mL
- (C) 0.5 mL
- (D) 10 mL
- (E) 25 mL
- (F) 50 mL
- (G) 250 mL

38. In the number 6,46,646, find the difference between the place values of the leftmost 6 and the rightmost 6. [\[Large Numbers\]](#)

- (A) 5,99,994
- (B) 6,00,006
- (C) 5,99,940
- (D) 5,99,400
- (E) 6,60,000
- (F) 594
- (G) 5,994

39. The summer monsoon brings heavy rain to India. Which description of these moisture-carrying winds is correct? [\[Winds, Storms & Cyclones\]](#)

- (A) They blow from the south-west, from land to sea
- (B) They blow from the south-west, from the sea onto the land
- (C) They blow from the east and carry no moisture
- (D) They blow from the north-west, from sea to land
- (E) They blow from the north-east, from land to sea
- (F) They blow straight downward from the sky
- (G) They blow from the south, from land to sea

40. A triangle has an area of 84 square cm and a height of 12 cm. What is the length of its base? [\[Perimeter & Area\]](#)

- (A) 21 cm
- (B) 28 cm
- (C) 96 cm
- (D) 168 cm
- (E) 7 cm
- (F) 12 cm
- (G) 14 cm

41. A boy's pulse is 80 beats per minute while resting and 120 beats per minute just after running. By what percentage did his pulse increase? [\[Transportation in Animals & Plants\]](#)

- (A) 60 percent
- (B) 25 percent
- (C) 150 percent
- (D) 40 percent
- (E) 33 percent
- (F) 50 percent
- (G) 20 percent

42. A water tank holds 12,500 litres. A town installs 1,250 such tanks. What is their total capacity in litres? [\[Large Numbers\]](#)

- (A) 1,37,50,000 litres
- (B) 1,56,25,00,000 litres
- (C) 1,56,25,000 litres
- (D) 15,62,50,000 litres
- (E) 15,625 litres
- (F) 15,62,500 litres
- (G) 1,42,50,000 litres

43. Storm winds blow very fast across the top of a tin roof, lowering the air pressure just above it. What is most likely to happen to the roof? [\[Winds, Storms & Cyclones\]](#)

- (A) The roof becomes heavier and sinks
- (B) The higher pressure below pushes the roof upward and can lift it off
- (C) The roof is pushed straight down even harder
- (D) The roof is pulled sideways into the wind
- (E) Air below escapes so the roof collapses inward
- (F) Nothing, because pressure is the same above and below
- (G) Higher pressure above presses the roof firmly down

44. A wheel rolls so that its circumference is 132 cm. Using pi as $\frac{22}{7}$, what is the diameter of the wheel? [\[Perimeter & Area\]](#)

- (A) 66 cm
- (B) 21 cm
- (C) 132 cm
- (D) 48 cm
- (E) 14 cm
- (F) 42 cm
- (G) 84 cm

45. A heart beats 70 times each minute. How many times does it beat in 100 days (take 1 day = 1440 minutes)? [\[Transportation in Animals & Plants\]](#)

- (A) 10080000 beats
- (B) 1080000 beats
- (C) 100800000 beats
- (D) 1008000 beats
- (E) 10008000 beats
- (F) 7200000 beats
- (G) 100800 beats

46. Round 38,472 and 51,649 each to the nearest thousand, and then add the two rounded numbers. [\[Large Numbers\]](#)

- (A) 88,000
- (B) 90,100
- (C) 90,000
- (D) 90,121
- (E) 91,000
- (F) 89,500
- (G) 89,000

47. A weather station records wind speeds over five hours as 40, 55, 60, 45 and 50 km per hour. What is the average wind speed? [\[Winds, Storms & Cyclones\]](#)

- (A) 250 km per hour
- (B) 48 km per hour
- (C) 60 km per hour
- (D) 50 km per hour
- (E) 45 km per hour
- (F) 55 km per hour
- (G) 52 km per hour

48. The length of a rectangle is three times its breadth, and the perimeter measures 64 cm. What is the area of this rectangle? [\[Perimeter & Area\]](#)

- (A) 48 square cm
- (B) 64 square cm
- (C) 96 square cm
- (D) 256 square cm
- (E) 192 square cm
- (F) 128 square cm
- (G) 384 square cm

49. Blood flows along a large artery at a steady speed of 30 cm per second. How far, in metres, does the blood travel in 2 minutes? [\[Transportation in Animals & Plants\]](#)

- (A) 60 metres
- (B) 6 metres
- (C) 3600 metres
- (D) 3.6 metres
- (E) 36 metres
- (F) 72 metres

50. How many thousands are there in 25 lakh? [\[Large Numbers\]](#)

- (A) 25
- (B) 2,500
- (C) 2,50,000
- (D) 250000
- (E) 5,000
- (F) 250
- (G) 25,000

51. A storm surge raises the sea by 4.5 m above the normal high tide of 1.5 m. How high is the water now, and how many times the normal tide is that? [\[Winds, Storms & Cyclones\]](#)

- (A) 6 m, which is 4.5 times the normal tide
- (B) 6 m, which is 6 times the normal tide
- (C) 3 m, which is 2 times the normal tide
- (D) 7.5 m, which is 5 times the normal tide
- (E) 6 m, which is 3 times the normal tide
- (F) 6 m, which is 4 times the normal tide
- (G) 4.5 m, which is 3 times the normal tide

52. A boy runs around a square park whose side is 75 m. How many complete rounds must he run to cover a total of 1.8 km? [\[Perimeter & Area\]](#)

- (A) 12 rounds
- (B) 3 rounds
- (C) 6 rounds
- (D) 8 rounds
- (E) 5 rounds
- (F) 60 rounds
- (G) 24 rounds

53. A potted plant is given 800 mL of water. Over one day it loses $\frac{5}{8}$ of this by transpiration. How much water remains in the plant and soil? [\[Transportation in Animals & Plants\]](#)

- (A) 350 mL
- (B) 320 mL
- (C) 500 mL
- (D) 250 mL
- (E) 200 mL
- (F) 480 mL
- (G) 300 mL

54. In the Indian system, what number comes just after 9,99,999? [\[Large Numbers\]](#)

- (A) 10,00,001
- (B) 10,00,000
- (C) 1,00,000
- (D) 99,99,999
- (E) 11,00,000
- (F) 99,90,000
- (G) 1,00,00,000

- 55.** The calm centre of a cyclone, called the eye, is roughly a circle of radius 15 km. Using pi as about 3.14, estimate the area of the eye. [\[Winds, Storms & Cyclones\]](#)
- (A) 225 square km
 - (B) 47.1 square km
 - (C) 70.65 square km
 - (D) 1,413 square km
 - (E) 706.5 square km
 - (F) 2,826 square km
- 56.** A field has an area of 2 hectares. Express this area in square metres (1 hectare = 10,000 square metres). [\[Perimeter & Area\]](#)
- (A) 20,000 square m
 - (B) 2,00,000 square m
 - (C) 20,00,000 square m
 - (D) 1,00,000 square m
 - (E) 2,500 square m
 - (F) 2,000 square m
 - (G) 20 square m
- 57.** In blood, for every 600 red blood cells there is about 1 white blood cell. In a sample containing 7200 red blood cells, how many white blood cells are expected? [\[Transportation in Animals & Plants\]](#)
- (A) 600 cells
 - (B) 120 cells
 - (C) 1200 cells
 - (D) 6 cells
 - (E) 72 cells
 - (F) 24 cells
 - (G) 12 cells
- 58.** A stadium can seat 1,05,000 people. On match day 87,650 people attended. How many seats stayed empty? [\[Large Numbers\]](#)
- (A) 27,350 seats
 - (B) 17,340 seats
 - (C) 16,350 seats
 - (D) 1,92,650 seats
 - (E) 17,450 seats
 - (F) 17,350 seats
- 59.** During a severe storm, 1,200 lightning flashes are recorded in 20 minutes. What is the average number of flashes per minute? [\[Winds, Storms & Cyclones\]](#)
- (A) 30 flashes per minute
 - (B) 1 flash per minute
 - (C) 24,000 flashes per minute
 - (D) 120 flashes per minute
 - (E) 6 flashes per minute
 - (F) 60 flashes per minute
 - (G) 600 flashes per minute

- 60.** A rectangular garden is 12 m by 8 m. A path 1 m wide runs inside it, all along the boundary. What is the area of this inner path? [\[Perimeter & Area\]](#)
- (A) 32 square m
 - (B) 60 square m
 - (C) 36 square m
 - (D) 20 square m
 - (E) 44 square m
 - (F) 40 square m
- 61.** A dialysis machine cleans 250 mL of a patient's blood every minute. How many litres of blood does it clean during a 4-hour session? [\[Transportation in Animals & Plants\]](#)
- (A) 36 litres
 - (B) 60000 litres
 - (C) 6 litres
 - (D) 1000 litres
 - (E) 600 litres
 - (F) 15 litres
 - (G) 60 litres
- 62.** Estimate the product 612×388 by first rounding each number to the nearest hundred. [\[Large Numbers\]](#)
- (A) 2,38,000
 - (B) 2,40,000
 - (C) 24,000
 - (D) 2,30,000
 - (E) 2,37,456
 - (F) 24,00,000
 - (G) 2,50,000
- 63.** Normal air pressure at sea level is about 1,01,325 pascals. Round this value to the nearest thousand. [\[Winds, Storms & Cyclones\]](#)
- (A) 1,01,000 pascals
 - (B) 1,00,000 pascals
 - (C) 1,01,325 pascals
 - (D) 1,01,400 pascals
 - (E) 1,10,000 pascals
 - (F) 1,01,300 pascals
 - (G) 1,02,000 pascals
- 64.** Find the perimeter of a semicircle (half-circle) whose radius is 7 cm. Use pi as $\frac{22}{7}$. [\[Perimeter & Area\]](#)
- (A) 22 cm
 - (B) 36 cm
 - (C) 25 cm
 - (D) 14 cm
 - (E) 29 cm
 - (F) 44 cm
 - (G) 18 cm

65. Each heartbeat makes two sounds, a lub and a dub. If the heart beats 74 times per minute, how many heart sounds are made in 5 minutes? [\[Transportation in Animals & Plants\]](#)

- (A) 370 sounds
- (B) 70 sounds
- (C) 222 sounds
- (D) 185 sounds
- (E) 148 sounds
- (F) 740 sounds
- (G) 1480 sounds

66. Which of these is the largest number: 89,99,999 or 9,81,000 or 98,01,000 or 9,80,999? [\[Large Numbers\]](#)

- (A) 98,01,000
- (B) 9,18,000
- (C) 9,08,000
- (D) 89,99,999
- (E) 8,91,000
- (F) 9,81,000
- (G) 9,80,999

67. A windmill turns 12 times each minute in a light breeze. During a storm it spins 3 times as fast. How many turns does it make in 10 minutes of storm? [\[Winds, Storms & Cyclones\]](#)

- (A) 36 turns
- (B) 180 turns
- (C) 3,600 turns
- (D) 300 turns
- (E) 360 turns
- (F) 90 turns
- (G) 120 turns

68. A right-angled triangle has its two shorter sides (the legs) measuring 9 cm and 12 cm. What is its area?

[\[Perimeter & Area\]](#)

- (A) 67.5 square cm
- (B) 45 square cm
- (C) 108 square cm
- (D) 54 square cm
- (E) 21 square cm
- (F) 27 square cm
- (G) 60 square cm

69. A plant transpires 90 mL of water during the 12 daylight hours. At night it transpires only one-third as fast per hour as the daytime rate. How much water does it lose in each night-time hour? [\[Transportation in Animals & Plants\]](#)

- (A) 5 mL per hour
- (B) 7.5 mL per hour
- (C) 2.5 mL per hour
- (D) 3.75 mL per hour
- (E) 30 mL per hour
- (F) 25 mL per hour
- (G) 1.5 mL per hour

- 70.** In the number 4,27,53,816, what is the sum of the digit in the lakhs place and the digit in the ten-lakhs place? [\[Large Numbers\]](#)
- (A) 27
 - (B) 12
 - (C) 7
 - (D) 9
 - (E) 11
 - (F) 13
 - (G) 6
- 71.** Warm air at 35 C rises and cools by about 6 C for every kilometre it climbs. What is its temperature after rising 4 km? [\[Winds, Storms & Cyclones\]](#)
- (A) 24 C
 - (B) 17 C
 - (C) 29 C
 - (D) 59 C
 - (E) 11 C
 - (F) 14 C
- 72.** A room measures 5 m by 4 m. Tiling the floor costs Rs 150 for each square metre. What is the total cost of tiling? [\[Perimeter & Area\]](#)
- (A) Rs 1,500
 - (B) Rs 2,700
 - (C) Rs 6,000
 - (D) Rs 750
 - (E) Rs 3,000
 - (F) Rs 300
 - (G) Rs 20
- 73.** Blood contains about 250,000 platelets in each cubic millimetre. In a 4 cubic millimetre sample, the number of platelets equals which power of ten? [\[Transportation in Animals & Plants\]](#)
- (A) 10 to the power 6
 - (B) 10 to the power 7
 - (C) 10 to the power 4
 - (D) 4×10 to the power 5
 - (E) 25×10 to the power 4
 - (F) 4×10 to the power 6
- 74.** A long book has 10,00,000 (ten lakh) words. If a reader manages 2,500 words a day, how many days are needed to finish it? [\[Large Numbers\]](#)
- (A) 4,000 days
 - (B) 800 days
 - (C) 40 days
 - (D) 4 days
 - (E) 2,500 days
 - (F) 250 days
 - (G) 400 days

75. Cyclone A has winds of 120 km per hour and Cyclone B has winds of 150 km per hour. By what percentage are Cyclone B's winds faster than Cyclone A's? [\[Winds, Storms & Cyclones\]](#)

- (A) 80 percent
- (B) 125 percent
- (C) 33 percent
- (D) 15 percent
- (E) 25 percent
- (F) 30 percent
- (G) 20 percent

76. A wire 48 cm long is bent to form a square. What area does the square enclose? [\[Perimeter & Area\]](#)

- (A) 576 square cm
- (B) 48 square cm
- (C) 288 square cm
- (D) 144 square cm
- (E) 96 square cm
- (F) 100 square cm

77. A child's heart beats 90 times a minute and an adult's beats 72 times a minute. Over 10 minutes, how many more times does the child's heart beat than the adult's? [\[Transportation in Animals & Plants\]](#)

- (A) 720 beats
- (B) 1620 beats
- (C) 18 beats
- (D) 160 beats
- (E) 180 beats
- (F) 90 beats

78. Round 47,62,380 to the nearest lakh. [\[Large Numbers\]](#)

- (A) 47,60,000
- (B) 50,00,000
- (C) 47,00,000
- (D) 47,62,000
- (E) 48,62,000
- (F) 4,80,00,000
- (G) 48,00,000

79. A weather centre tracks a cyclone for 3 full days, after which it sits over the coast for another 18 hours. What is the total tracking time in hours? [\[Winds, Storms & Cyclones\]](#)

- (A) 72 hours
- (B) 96 hours
- (C) 54 hours
- (D) 78 hours
- (E) 90 hours
- (F) 42 hours

80. A circular flower bed has a radius of 21 m and is to be fenced at Rs 10 per metre. Using pi as $\frac{22}{7}$, what is the cost of the fence? [\[Perimeter & Area\]](#)

- (A) Rs 660
- (B) Rs 1,386
- (C) Rs 1,300
- (D) Rs 13,860
- (E) Rs 2,640
- (F) Rs 1,380
- (G) Rs 1,320

81. Blood takes about 20 seconds to make one complete trip around the body. How many complete trips does the blood make in one hour? [\[Transportation in Animals & Plants\]](#)

- (A) 18 trips
- (B) 3600 trips
- (C) 30 trips
- (D) 180 trips
- (E) 120 trips
- (F) 60 trips
- (G) 200 trips

82. Read the number 50,05,005. Which words describe it correctly? [\[Large Numbers\]](#)

- (A) Fifty lakh five hundred five
- (B) Fifty lakh five thousand fifty
- (C) Fifty lakh fifty thousand five
- (D) Fifty lakh fifty thousand fifty
- (E) Five lakh five thousand five
- (F) Five crore five thousand five
- (G) Fifty lakh five thousand five

83. Floodwater left by a storm fills a basement with 60,000 litres. A pump removes 500 litres each minute. How many minutes does it take to clear the water? [\[Winds, Storms & Cyclones\]](#)

- (A) 240 minutes
- (B) 120 minutes
- (C) 12 minutes
- (D) 60 minutes
- (E) 20 minutes
- (F) 1,200 minutes

84. A rectangular lawn 14 m by 10 m has a semicircular flower bed of radius 7 m attached along one of its 14 m sides. What is the total area? Use pi as $\frac{22}{7}$. [\[Perimeter & Area\]](#)

- (A) 203 square m
- (B) 140 square m
- (C) 217 square m
- (D) 154 square m
- (E) 231 square m
- (F) 77 square m

- 85.** Inside a stem, water rises by root pressure at a steady 2.5 cm every minute. How many minutes does it take for the water to rise a full 1 metre up the stem? [\[Transportation in Animals & Plants\]](#)
- (A) 25 minutes
 - (B) 2.5 minutes
 - (C) 40 minutes
 - (D) 20 minutes
 - (E) 400 minutes
 - (F) 4 minutes
- 86.** A sweet shop has 84,000 pencils to pack, with 24 pencils going into each box. How many boxes are filled? [\[Large Numbers\]](#)
- (A) 35,000 boxes
 - (B) 3,600 boxes
 - (C) 7,000 boxes
 - (D) 350 boxes
 - (E) 3,500 boxes
 - (F) 3,000 boxes
- 87.** During a storm the steady wind blows at 40 km per hour while sudden gusts reach 70 km per hour. What is the ratio of gust speed to steady speed in its simplest form? [\[Winds, Storms & Cyclones\]](#)
- (A) 40 to 70
 - (B) 7 to 3
 - (C) 7 to 4
 - (D) 3 to 4
 - (E) 7 to 40
 - (F) 2 to 1
 - (G) 4 to 7
- 88.** A park measures 50 m by 30 m and has a path 2 m wide all around its outside. Gravelling the path costs Rs 20 per square metre. What is the total cost? [\[Perimeter & Area\]](#)
- (A) Rs 6,620
 - (B) Rs 30,000
 - (C) Rs 7,200
 - (D) Rs 6,600
 - (E) Rs 6,700
 - (F) Rs 336
 - (G) Rs 6,720
- 89.** A person loses about 1.5 litres of water as sweat and 1 litre as urine in a day, with total water loss of 5 litres. What fraction of the total water loss is sweat? [\[Transportation in Animals & Plants\]](#)
- (A) $\frac{3}{10}$
 - (B) $\frac{2}{5}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{1}{2}$
 - (E) $\frac{3}{5}$
 - (F) $\frac{1}{4}$
 - (G) $\frac{1}{5}$

90. What is the value of one crore minus one? [\[Large Numbers\]](#)

- (A) 1,00,000
- (B) 9,99,99,999
- (C) 99,99,990
- (D) 9,99,999
- (E) 1,00,00,001
- (F) 99,99,999
- (G) 9,99,99,990

91. A barometer measures air pressure. When its reading falls steadily over a few hours, what kind of weather does this most often warn is coming? [\[Winds, Storms & Cyclones\]](#)

- (A) Clear, calm and sunny weather is coming
- (B) Colder but dry and perfectly still air
- (C) Brighter sunshine and gentle winds
- (D) A long, dry, rainless spell ahead
- (E) Rising temperatures with no clouds
- (F) No change in the weather at all
- (G) Stormy, rainy weather is likely approaching

92. A square's side is increased from 10 cm to 15 cm. By how much does its area increase? [\[Perimeter & Area\]](#)

- (A) 125 square cm
- (B) 25 square cm
- (C) 225 square cm
- (D) 50 square cm
- (E) 5 square cm
- (F) 100 square cm

93. Each heartbeat pumps 80 mL of blood. How many beats are needed to pump a total of 2 litres (2000 mL) of blood? [\[Transportation in Animals & Plants\]](#)

- (A) 25 beats
- (B) 250 beats
- (C) 20 beats
- (D) 16 beats
- (E) 40 beats
- (F) 160 beats

94. In the number 8,53,672, the digit 5 sits in which place, and what is its place value? [\[Large Numbers\]](#)

- (A) Lakhs place, value 5,00,000
- (B) Ones place, value 5
- (C) Ten-thousands place, value 50,000
- (D) Thousands place, value 5,000
- (E) Hundreds place, value 500
- (F) Hundreds place, value 50,000
- (G) Ten-lakhs place, value 50,00,000

- 95.** A cyclone is first spotted 540 km away, moving steadily toward the coast at 20 km per hour. How far away will it be after 6 hours? [\[Winds, Storms & Cyclones\]](#)
- (A) 400 km
 - (B) 420 km
 - (C) 660 km
 - (D) 540 km
 - (E) 90 km
 - (F) 120 km
 - (G) 480 km
- 96.** A triangle is drawn inside a rectangle of size 16 cm by 9 cm, sharing the same 16 cm base and the full 9 cm height. What is the triangle's area, and what fraction of the rectangle is it? [\[Perimeter & Area\]](#)
- (A) 144 square cm, which is half the rectangle
 - (B) 36 square cm, which is one-quarter of the rectangle
 - (C) 72 square cm, which is one-third of the rectangle
 - (D) 108 square cm, which is half the rectangle
 - (E) 144 square cm, which is the whole rectangle
 - (F) 72 square cm, which is one-quarter of the rectangle
 - (G) 72 square cm, which is half the rectangle
- 97.** A nurse counts 27 pulse beats in 20 seconds. What is the patient's heart rate per minute? [\[Transportation in Animals & Plants\]](#)
- (A) 27 beats per minute
 - (B) 90 beats per minute
 - (C) 540 beats per minute
 - (D) 81 beats per minute
 - (E) 54 beats per minute
 - (F) 60 beats per minute
 - (G) 108 beats per minute
- 98.** A car costs Rs 6,75,000. A company buys 18 such cars for its staff. What is the total cost? [\[Large Numbers\]](#)
- (A) Rs 6,75,018
 - (B) Rs 12,15,00,000
 - (C) Rs 12,15,000
 - (D) Rs 1,35,00,000
 - (E) Rs 1,20,50,000
 - (F) Rs 1,21,50,000
- 99.** Moving air pushes with a force of about 10 newtons on each square centimetre. What total force does the wind push on a window measuring 50 cm by 40 cm? [\[Winds, Storms & Cyclones\]](#)
- (A) 900 newtons
 - (B) 20,000 newtons
 - (C) 20,00,000 newtons
 - (D) 18,000 newtons
 - (E) 5,000 newtons
 - (F) 2,00,000 newtons
 - (G) 2,000 newtons

100. A wire is bent to form a rectangle 14 cm by 8 cm. It is then straightened and bent into a circle. Using π as $\frac{22}{7}$, what is the radius of the circle? [\[Perimeter & Area\]](#)

- (A) 14 cm
- (B) 3.5 cm
- (C) 44 cm
- (D) 11 cm
- (E) 7 cm
- (F) 22 cm
- (G) 8 cm